Introduction

Objectives of the Study :

The main objective of this study is to analyze and classify young professional football players based on their statistical performance. The analysis focuses on U23 players from several professional leagues (Ligue 2, Championship, Segunda division, Serie B) and aims to identify different performance profiles among attackers, midfielders, and defenders. By applying PCA and clustering techniques, the study seeks to group players with similar statistical characteristics and highlight the diversity of playing styles among young talents.

Data-Driven Player Analysis in Modern Football :

Data-driven player analysis relies on quantitative match statistics to evaluate performance and identify patterns in player behavior. Techniques such as PCA allow the reduction of complex datasets into a few main performance dimensions, while clustering methods group players according to similar statistical profiles. These approaches make it easier to compare players and identify typical performance archetypes within different positions.

Structure of the Report :

This report is structured in several parts. After presenting the methodology and dataset, the analysis examines attacking players, midfielders, and defenders separately. For each position, the main performance dimensions and player clusters are identified and illustrated through individual player analyses. The report concludes with a discussion of the strengths and limitations of the model and its potential applications for football analysis and scouting.

Data and Methodology

Dataset Description :

The dataset used in this study was built from match-event statistics collected for young professional football players competing in several European second-tier leagues. The analysis focuses on players from the Championship, Ligue 2, Segunda División, and Serie B, providing a cross-league perspective on player performance and role diversity.

For each player, the dataset includes a wide range of variables related to offensive production, passing activity, defensive contribution, duels, dribbling, and disciplinary actions. These raw statistics were then converted into per-90-minute

indicators in order to improve comparability between players with different amounts of playing time.

The dataset was subsequently divided by position into three subgroups: attackers, midfielders, and defenders, allowing each role to be analyzed separately according to the statistical variables most relevant to its tactical function.

Player Selection Criteria (U23 Players) :

The study focuses exclusively on under-23 players, with the aim of identifying promising young talents and comparing emerging profiles across leagues. In order to ensure a sufficient level of statistical reliability, only players with a meaningful amount of playing time were retained in the final sample.

This filtering step is essential because match statistics can be highly unstable for players with very limited minutes. Restricting the sample to U23 players with sufficient exposure makes it possible to reduce statistical noise and produce more robust player comparisons.

Statistical Indicators Used :

Different sets of variables were selected depending on the position analyzed:

goals per 90 minutes, shots per 90 minutes, shots on target per 90 minutes, key passes per 90 minutes, successful dribbles per 90 minutes, duels won per 90 minutes, assists per 90 minutes, key passes per 90 minutes, successful dribbles per 90 minutes, passes per 90 minutes, interceptions per 90 minutes, duels won per 90 minutes, tackles per 90 minutes, interceptions per 90 minutes, duels won per 90 minutes, passes per 90 minutes.

These variables were chosen because they summarize the main dimensions of performance for each role while remaining sufficiently comparable across leagues.

Data Normalization and Shrinkage:

Before applying dimensionality reduction and clustering techniques, the data were processed in two main steps: shrinkage and standardization.

First, a shrinkage procedure was applied to the per-90 statistics in order to reduce the instability of extreme values, particularly for players with fewer minutes. This approach pulls individual values toward the average, making the estimates more robust and limiting the influence of statistical outliers.

In this study, shrinkage was applied separately by league and by positional group. This means that each player's statistics were adjusted relative to the average performance of players in the same league and playing the same position. This

approach helps control for structural differences between leagues and ensures that players are compared within similar competitive contexts.

The shrunk statistic for player i is calculated as a weighted average between the player's observed value and the group mean:

$$x_i^{shrunk} = \frac{n_i}{n_i + k} \cdot x_i + \frac{k}{n_i + k} \cdot \bar{x}_{group}$$

where:

- x_i is the observed per-90 statistic for player i
- $\bar{x}_{group}$ is the average value of the statistic for players in the **same league and position**
- n_i represents the amount of playing time for player i
- k is a smoothing parameter controlling the strength of the shrinkage.

Second, all shrunk variables were standardized using z-score normalization. This transformation ensures that each variable has a comparable scale and allows the PCA to reflect relative variation rather than absolute differences in units.

The final variables used in the analysis are therefore the standardized shrunk indicators, noted in the dataset with the prefix z_ and the suffix _shrunk.

<u>Principal Component Analysis (PCA):</u>

A Principal Component Analysis (PCA) was performed separately for attackers, midfielders, and defenders using the FactoMineR package in R. The purpose of this step was to reduce the dimensionality of the data and identify the main statistical dimensions structuring player performance within each position.

For each positional group, the PCA was conducted on the selected standardized variables only. The retained components were then interpreted using:

eigenvalues and explained variance, variable contributions, variable-axis correlations, graphical tools such as scree plots and correlation circles.

The number of principal components retained followed a simple rule:

If the first two components explained at least 70% of cumulative variance and the third component had an eigenvalue below 1, then 2 axes were retained;

otherwise, 3 axes were kept.

This rule made it possible to balance interpretability and explanatory power across positions.

Construction of Position-Specific Performance Scores:

To facilitate the comparison of players within each positional group, three synthetic performance indicators were constructed: score_attack, score_midfield, and score_defense. These scores summarize each player's statistical contribution by combining several key performance indicators relevant to their position.

General Formulation

The three performance scores are calculated as **weighted sums of the shrunk and standardized variables**:

$$Score_i = \sum_{k=1}^{K} w_k \cdot z_{ik}$$

where:

- w_k represents the **weight assigned to variable** k
- z_{ik} represents the **standardized value (z-score) of variable** k **for player** i

In other words, each score corresponds to a **weighted linear combination of the standardized performance indicators**. This approach allows different statistical variables to contribute unequally to the final score, depending on their relative importance for the position being analyzed.

By combining these weighted standardized indicators, the resulting scores provide a synthetic measure of statistical impact within each positional group. Because the variables are standardized, a positive score indicates performance above the dataset average, while a negative score reflects below-average statistical contribution. These scores are therefore useful for quickly identifying players with strong statistical profiles, while remaining consistent across different performance dimensions.

Clustering Methodology:

After the PCA, clustering was performed on the retained principal components in order to identify groups of players with similar statistical profiles.

The clustering procedure used the k-means algorithm, with the number of clusters selected through a combination of:

the within-cluster sum of squares (WSS) elbow method and the silhouette method

The final number of clusters retained for each position was 3 clusters for attackers, 2 clusters for midfielders and 6 clusters for defenders.

The clustering was performed on the PCA coordinates rather than on the raw variables, which helps reduce noise and improves the separation of profiles in a lower-dimensional space.

This combined PCA + k-means framework allows the model to identify general player archetypes based on statistical similarity, while also providing a more

interpretable structure for comparing young players across different leagues and positions.

Player Analysis

1) Attackers Analysis

Performance Dimensions :

The Principal Component Analysis (PCA) conducted on attacking players identified two main dimensions structuring their performance profiles.

The first principal component (PC1) is strongly associated with variables related to finishing ability and shooting activity. Goals per 90 minutes, shots per 90 minutes, and shots on target per 90 minutes are the variables contributing most strongly to this axis. As such, PC1 can be interpreted as a dimension capturing offensive productivity and efficiency in front of goal.

The second principal component (PC2) reflects a different aspect of attacking performance. It is primarily driven by variables related to successful dribbles, key passes, and duels won. This axis captures a player's creative contribution and attacking dynamism, highlighting those who are able to progress the ball, create opportunities, and destabilize defensive structures.

Taken together, these two dimensions provide a simplified but meaningful representation of attacking performance, distinguishing players who specialize in finishing from those who contribute more strongly to chance creation and offensive progression.

Attacking Player Profiles:

Based on the PCA results, a clustering analysis was performed in order to identify groups of attackers with similar statistical characteristics. The model identified three distinct attacking profiles, each corresponding to a different offensive role.

Cluster 1 – Balanced Attackers

Players in this cluster display moderate values on both principal components. They do not appear to specialize strongly in either finishing or creativity, but rather exhibit a balanced offensive profile.

From a statistical perspective, these players generally combine moderate levels of goals, shots, dribbles, and key passes. Their contribution is therefore distributed across several offensive actions rather than concentrated in a single dimension of play.

From a tactical point of view, this cluster may represent versatile attackers capable of contributing to different phases of offensive play. They can participate in build-up phases, support chance creation, and finish actions when required, making them adaptable to a variety of attacking systems.

Cluster 2 – Finishing Specialists

The second cluster is characterized by high values on the finishing dimension (PC1) and more neutral values on the creativity dimension.

Players in this group show strong goal-scoring productivity and a relatively high volume of shots per match. Their statistical profiles indicate that they are heavily involved in the final phase of attacking sequences and are primarily responsible for converting chances.

In footballing terms, these players correspond to finishing-oriented forwards, often positioned close to the goal and mainly focused on scoring rather than chance creation.

Cluster 3 – Creative Attackers

Players in this cluster display high values on the creativity dimension (PC2) and lower or more moderate values on the finishing dimension.

Statistically, these players are characterized by high dribbling output, a significant number of key passes, and strong involvement in offensive duels.

This profile suggests attackers who contribute mainly to chance creation and attacking progression, rather than focusing primarily on finishing.

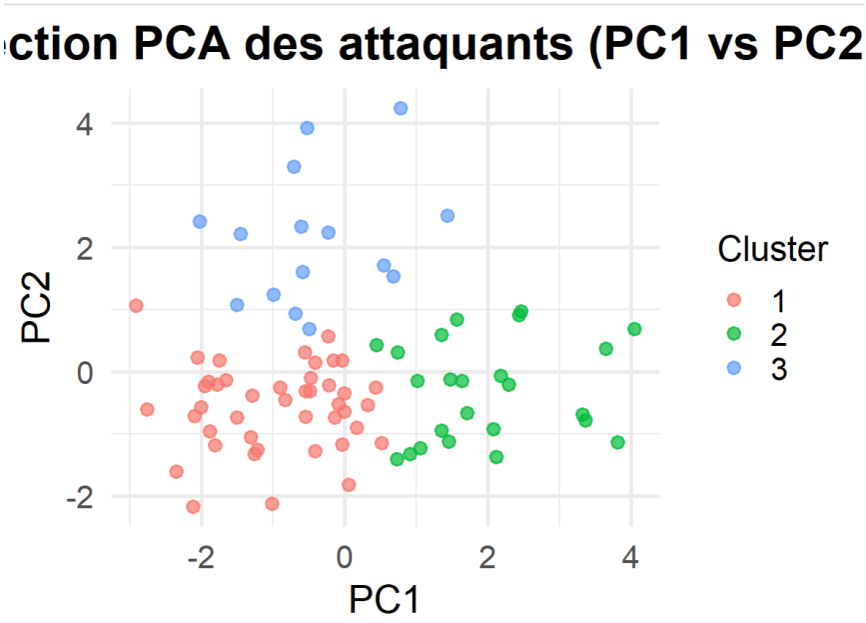
From a tactical perspective, these players often operate as creative forwards, wide attackers, or second strikers, capable of destabilizing defensive lines through dribbling and playmaking.

<u>Methodological Limitations:</u>

Several limitations should nevertheless be acknowledged when interpreting the attacking profiles. First, attackers often display hybrid statistical profiles, combining both finishing ability and creative contribution. This can make their classification within a single cluster less intuitive, as some players may not fit perfectly into a clearly defined category.

In addition, the model relies exclusively on quantitative match statistics, which do not capture important dimensions of performance such as off-ball movement, tactical positioning, or role-specific instructions within a team.

As a result, the clustering procedure identifies statistical similarity rather than exact tactical roles, which may sometimes limit the interpretation of attacking profiles.

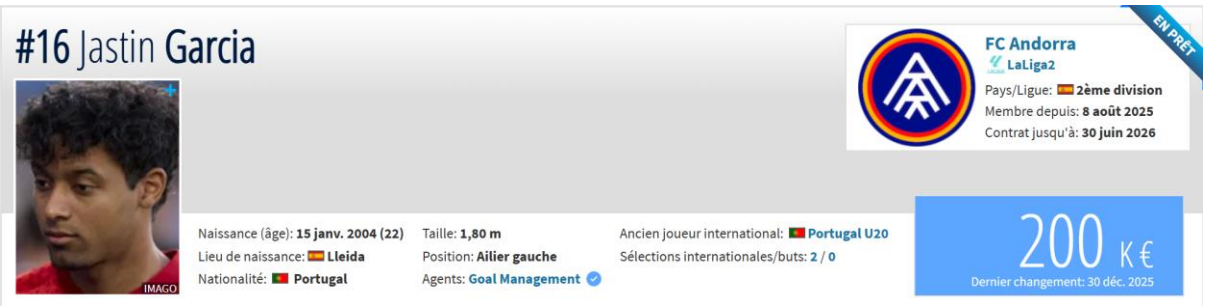


Example:

	player_name	source_league_name	age	score_attack	cluster
1	Dani Díaz	Segunda División	19	1.0306039	1
2	Chupe	Segunda División	21	0.8832100	1
3	T. Marras	Serie B	21	0.8823776	1
4	J. Makama	Championship	21	0.8514023	1
5	Adrián Niño Heredia	Segunda División	21	0.8380999	1
6	A. Cissè	Serie B	19	0.6677951	1
7	F. Abiuso	Serie B	22	0.6417434	1
8	A. Raimondo	Serie B	21	0.5780159	1
9	Antonino	Segunda División	21	1.2326929	2
10	O. Maamma	Championship	20	0.9346370	2
11	S. Shpendi	Serie B	22	0.7117874	2
12	Minsu Kim	Segunda División	19	0.6065743	2
13	Marcos Fernández	Segunda División	22	0.4742495	2
14	R. Burnete	Serie B	21	0.1853237	2
15	Jorge Pascual	Segunda División	22	0.1687740	2
16	O. Sadik	Ligue 2	21	0.1494250	2
17	Jeremy Arévalo	Segunda División	20	1.5841628	3
18	C. Shpendi	Serie B	22	0.9477160	3
19	L. Stassin	Ligue 2	21	0.8201587	3
20	N. Irankunda	Championship	19	0.6022067	3
21	Jay Robinson	Championship	18	0.4462574	3
22	Gorka Carrera	Segunda División	20	0.4018735	3
23	Thalys	Segunda División	20	-0.1024322	3

24	V. Ouotro	Ligue 2	20	-0.2237553	3

Individual Analysis – Antonino (Jastin Garcia)



Antonino is a 21-year-old Portuguese attacking player currently playing for FC Andorra in the Spanish Segunda División. During the 2025 season, he accumulated 965 minutes across 21 appearances, indicating a relatively regular level of involvement within the squad.

From a statistical perspective, Antonino displays a dynamic and versatile attacking profile, characterized by strong involvement across several offensive dimensions.

His shooting activity is relatively high, with 2.15 shots per 90 minutes and 1.40 shots on target per 90 minutes, both clearly above the dataset averages. These indicators suggest a player who is able to generate shooting opportunities consistently and occupy dangerous attacking zones.

Beyond finishing, Antonino stands out in ball progression and offensive duels. He records 3.45 successful dribbles per 90 minutes, almost three times the dataset average, while also attempting 6.53 dribbles per 90 minutes. This suggests a player who actively challenges defenders and attempts to destabilize opposition structures through individual actions.

He also shows a strong physical presence in attacking duels, averaging 16.97 duels per 90 minutes and winning more than 9, which indicates a forward capable of competing physically with defenders and retaining possession in contested situations.

His contribution to chance creation is also notable, with 1.31 key passes per 90 minutes, above the dataset average of 0.99. This indicates that he contributes not only through shooting and dribbling, but also through the generation of opportunities for teammates.

Overall, Antonino's statistical profile reflects a multi-dimensional attacking player, combining shooting activity, dribbling ability, physical engagement, and creative contribution.

Although Antonino belongs to the finishing-oriented cluster, his individual statistics also reveal strong creative and ball progression abilities.

Individual Analysis – Nestory Irankunda

Nestory Irankunda is a 19-year-old Australian right winger currently playing for Watford in the English Championship. During the 2025 season, he accumulated 1,287 minutes across 29 appearances, reflecting a consistent level of involvement despite alternating between starts and substitute appearances.

Statistically, Irankunda's profile clearly leans toward attacking creativity and offensive dynamism.

He averages 1.68 key passes per 90 minutes, a value well above the dataset average, highlighting a strong capacity to create chances. This suggests a player who is actively involved in attacking build-up and regularly generates opportunities for teammates.

Irankunda also stands out through his dribbling activity and ball progression. He records 4.20 dribble attempts per 90 minutes and 2.17 successful dribbles per 90 minutes, indicating a player who frequently takes on defenders in one-on-one situations and helps advance the ball into dangerous areas.

In addition, he shows a strong physical presence in attacking duels, averaging 16.64 duels per 90 minutes and winning 8.32, which reflects significant involvement in contested attacking situations.

However, his goal-scoring output is more moderate, with 0.14 goals per 90 minutes, below the dataset average. This suggests that his main contribution lies less in finishing and more in chance creation and progression.

Overall, Irankunda's profile corresponds closely to that of a creative and energetic attacking player, whose influence extends beyond pure goal scoring.

Comparison of Profiles:

The comparison between Antonino and Irankunda highlights two distinct offensive styles. Antonino presents a more balanced profile, combining shooting volume, dribbling activity, and physical engagement in duels. In contrast, Irankunda stands out more clearly through his creative contribution, particularly in terms of key passes and ball progression through dribbling.

These two examples illustrate the diversity of attacking roles identified among young players in the dataset.

Clarification on Cluster Representation:

It is also important to clarify why Irankunda was selected as the illustrative example for Cluster 3 rather than players such as Jeremy Arévalo. Although Arévalo also belongs to this cluster, his statistical profile is largely driven by strong finishing indicators, whereas his creative metrics remain more moderate. Irankunda therefore offers a more representative example of the creative attacking profile identified by the clustering model.

Conclusion:

Despite the limitations associated with using only match statistics, the model provides a structured and robust framework for identifying player profiles. It offers a clear representation of offensive playing styles and facilitates the comparison of young players across multiple leagues.

More broadly, this approach highlights the diversity of attacking roles in modern football and demonstrates the value of data-driven methods for understanding player performance and stylistic differences.

2) Midfielders Analysis

Performance Dimensions:

The first principal component (PC1) primarily contrasts defensive involvement with offensive playmaking activity. This dimension shows a strong positive correlation with interceptions per 90 minutes (0.83) and passes per 90 minutes (0.52), while displaying negative correlations with key passes (-0.75) and assists (-0.56). As a result, PC1 can be interpreted as a dimension separating defensive or possession-oriented midfielders from more attack-minded playmakers who contribute directly to chance creation.

The second principal component (PC2) reflects the level of physical and dynamic engagement in midfield actions. It is mainly driven by successful dribbles (0.75) and duels won (0.74). Players with high scores on this dimension tend to be more active in one-on-one situations and physical contests in midfield, indicating a more dynamic and combative playing style.

The third principal component (PC3) captures a dimension associated with ball circulation and offensive support. The strongest correlations on this axis come from passes per 90 minutes (0.66) and assists (0.46). This dimension highlights midfielders who are strongly involved in the distribution of the ball and in supporting attacking phases.

Together, these three components illustrate the diversity of roles within midfield positions, distinguishing between defensive midfielders, physically active box-to-box players, and creative playmaking profiles.

Midfielder Player Profiles:

The clustering analysis performed on midfield players reveals two main statistical profiles within the dataset.

Cluster 1 – Offensive/ build-up Miedfielders

Cluster 1, which contains the majority of players (76 midfielders), is characterized by a negative average value on PC1 (-1.13), while PC2 and PC3 remain close to zero. Since PC1 contrasts defensive contribution with offensive playmaking indicators, players in this cluster tend to display stronger involvement in offensive creation, particularly through key passes and assists. These midfielders can therefore be interpreted as creative or attacking-oriented profiles, whose primary contribution lies in chance creation and offensive build-up.

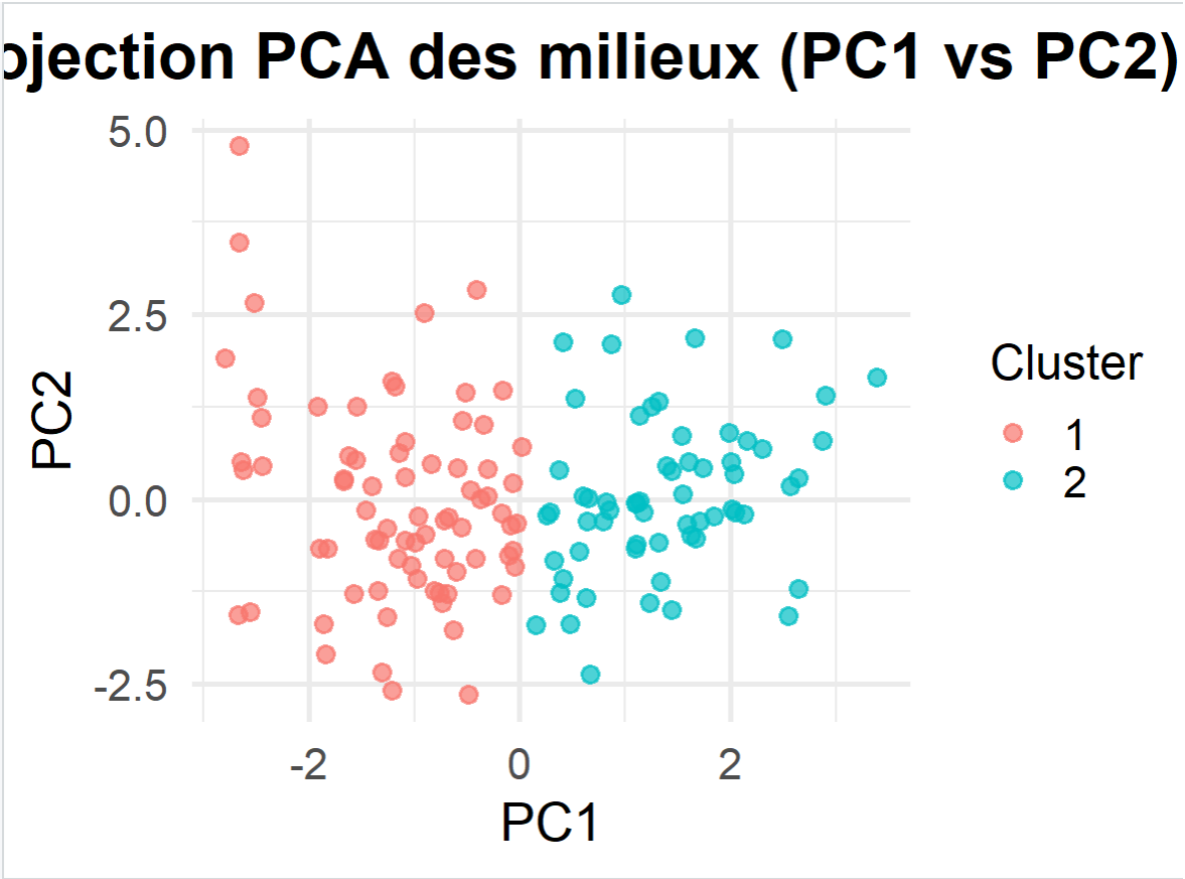
Cluster 2 – defensive/possession-oriented Miedfielders

Cluster 2, composed of 62 players, displays a strong positive average value on PC1 (1.39). Given the structure of PC1, this suggests midfielders who are more involved in defensive actions and ball recovery, such as interceptions and physical duels, while also contributing to ball circulation through passing volume. These players therefore correspond more closely to defensive or possession-oriented midfielders, often operating in deeper positions within the team's tactical structure.

Methodological limitations:

However, it remains possible to identify players with box-to-box tendencies within a given cluster. This is particularly the case for players displaying high values on PC2, reflecting strong involvement in duels and dribbling, combined with intermediate values on PC1, which indicates a balance between defensive contribution and attacking participation.

In other words, the box-to-box profile does not emerge as an independent cluster in this analysis. Instead, it appears as an individual sub-profile, observable through a player's position within the PCA space rather than as a clearly separated statistical category.



Example:

	player_name	age	score_midfield	cluster
1	Gessime Yassine	20	1.7302976	1
2	S. Pafundi	19	1.0333772	1
3	A. Fatawu	21	1.0057725	1
4	R. Lutin	22	0.9983325	1
5	M. Adeline	22	0.9776858	1
6	Peio Canales	20	0.8246256	1
7	E. Koné	19	0.7704264	1
8	I. Miladinović	22	0.7495801	1
9	M. Detourbet	18	1.2622726	2
10	E. Zabi	19	0.8265468	2
11	S. Faye	22	0.5856822	2
12	Jacob Wright	20	0.4645951	2
13	H. Amass	18	0.4263821	2

14	Aleix Garrido	21	0.4240762	2
15	L. Hasa	21	0.3991143	2
16	T. Devlin	22	0.3904698	2

Individual Analysis – Gessime Yassine

Gessime Yassine is a 20-year-old French midfielder currently playing for Strasbourg (played at Dunkerque ligue 2 first half of the season) he accumulated 803 minutes across 13 appearances, with a strong average rating of 7.08, suggesting a consistently high level of performance when involved.

From a creative perspective, Yassine stands out as one of the most productive midfielders in the dataset. His 0.67 assists per 90 minutes is extremely high compared to the dataset average of 0.11, indicating a player who plays a major role in chance creation and final passes in attacking sequences. His 1.34 key passes per 90 minutes further confirms his significant involvement in offensive playmaking.

Yassine also demonstrates strong ability in ball progression and one-on-one situations. He attempts 5.04 dribbles per 90 minutes and successfully completes 2.47, both clearly above the dataset averages of 2.11 and 0.91 respectively. These numbers suggest a midfielder capable of breaking defensive lines and advancing the ball through individual actions, a valuable quality in attacking transitions.

In addition to his creative contribution, Yassine also shows a high level of physical engagement in midfield duels. He participates in 16.36 duels per 90 minutes and wins 8.74, both significantly above the dataset averages of 11.20 and 5.33. These statistics indicate a midfielder who actively competes in contested situations and contributes to maintaining possession or recovering the ball in midfield battles.

Yassine's statistical profile corresponds to that of a dynamic and highly creative attacking midfielder. His capacity to influence both the progression and final phases of attacking sequences makes him a particularly impactful player despite his relatively limited playing time.

Individual Analysis – Jacob Wright

Jacob Wright is a 20-year-old English midfielder currently playing for Norwich City in the English Championship. During the 2025 season he accumulated 1,037 minutes across 21 appearances, including 11 starts, indicating a rotational yet regular role within the squad.

From a statistical perspective, Wright's profile is strongly oriented toward ball circulation and midfield control. He averages 45.04 passes per 90 minutes, significantly above the dataset average of approximately 31 passes per 90 minutes, reflected in a positive standardized score ($z = 0.82$). This suggests a midfielder who is heavily involved in possession phases and distribution of the ball.

Wright also contributes to chance creation, producing 1.91 key passes per 90 minutes, well above the dataset average of around 1.08, with a strong standardized value ($z = 1.65$). This indicates that he contributes not only to maintaining possession but also to the progression of attacking sequences.

Defensively, Wright demonstrates a solid level of activity in ball recovery. He records 2.43 tackles per 90 minutes, clearly above the dataset average of approximately 1.83, suggesting active involvement in regaining possession. His 0.78 interceptions per 90 minutes remain close to the dataset average, reinforcing the idea of a midfielder contributing to defensive stability.

However, Wright's attacking output in terms of finishing remains relatively limited. He records no goals and only 0.78 shots per 90 minutes, both below the dataset averages. Furthermore, his dribbling activity remains relatively modest compared to more offensively oriented midfield profiles.

Overall, Wright's statistical profile corresponds to that of a possession-oriented central midfielder, combining high passing volume, solid defensive activity, and moderate chance creation. His role appears primarily focused on organizing play and

maintaining ball circulation, rather than directly influencing attacking phases through scoring or individual dribbling actions.

Comparative Interpretation:

The individual analyses of Gessime Yassine and Jacob Wright illustrate the diversity of profiles identified within the midfield dataset. While Yassine stands out for his strong creative contribution, particularly through assists, dribbling, and offensive involvement, Wright displays a profile more oriented toward ball circulation and midfield organization, characterized by high passing volume and solid defensive contribution.

Together, these examples highlight how midfield roles can vary significantly, ranging from creative attacking influences to players primarily responsible for structuring possession and maintaining balance in midfield.

Clarification of the Clustering Structure:

The two clusters identified for midfielders represent statistical tendencies rather than strict tactical roles. Cluster 1 mainly groups players involved in offensive creation (assists and key passes), while Cluster 2 includes midfielders more associated with defensive contribution and ball circulation (interceptions and passing volume). Hybrid roles such as box-to-box midfielders do not appear as separate clusters but rather as intermediate profiles within the PCA space.

Conclusion:

Overall, the analysis highlights the diversity of midfield roles within the dataset. The PCA identifies three key dimensions of performance: defensive contribution, physical involvement, and ball circulation.

The clustering results broadly distinguish creative attacking midfielders from defensive or possession-oriented midfielders. However, midfield roles often exist along a continuum, with hybrid profiles combining multiple types of contribution.

Despite these nuances, the model provides a useful framework for identifying midfield profiles and comparing players across leagues, offering insights into how young midfielders contribute to different phases of play.

3) Defenders Analysis

Performance Dimensions:

The Principal Component Analysis (PCA) conducted on defenders identifies two main performance dimensions structuring the statistical profiles of defensive players.

The first principal component (PC1) is primarily driven by tackles per 90 minutes, and duels won per 90 minutes, both strongly positively correlated with this axis. This dimension captures the level of direct defensive engagement, reflecting a defender's ability to challenge opponents and recover possession through tackles and physical duels. Players with high PC1 values therefore tend to be more involved in physical defensive confrontations and ball-winning situations.

The second principal component (PC2) is mainly associated with interceptions per 90 minutes and passes per 90 minutes. This axis reflects a different defensive profile, related to positional awareness and ball distribution. Players with high PC2 values tend to anticipate passes, intercept opposition plays, and contribute to the circulation of possession after ball recovery.

Together, these two dimensions highlight two complementary aspects of defensive performance: direct defensive engagement through tackles and duels, and positional defensive intelligence combined with ball distribution.

Defender Player Profiles:

The clustering analysis identified six distinct statistical profiles among defenders, reflecting different types of defensive contribution within the dataset.

Cluster 1 – Physical Ball-Winning Defenders

Players in this cluster display high values on PC1, but relatively low values on PC2. This indicates strong involvement in defensive duels and tackles, combined with lower participation in ball circulation and interception-based defending. These defenders tend to focus on direct defensive engagement, often confronting opponents in one-on-one situations.

Cluster 2 – Balanced Defensive Profiles

Cluster 2 contains players with values close to zero on both axes, suggesting defenders with balanced statistical profiles. These players contribute moderately to both defensive actions and ball circulation, representing versatile defensive profiles without strong specialization.

Cluster 3 – Dominant Defensive Enforcers

This cluster shows the highest values on PC1, indicating defenders heavily involved in physical defensive actions, particularly tackles and duels. These players act as defensive enforcers, whose primary role is to stop attacking actions and recover possession through direct defensive pressure.

Cluster 4 – Low-Activity Defensive Profiles

Players in this cluster display negative values on both axes, indicating lower involvement in both physical defensive actions and ball circulation. These defenders may occupy more situational or supporting roles, with limited statistical impact in the variables considered.

Cluster 5 – Ball-Playing Defenders

Cluster 5 presents the highest values on PC2, indicating defenders who combine interceptions with strong passing involvement. These players contribute significantly to building play from the back and maintaining possession, corresponding to modern ball-playing defensive profiles.

Cluster 6 – Positional Defensive Profiles

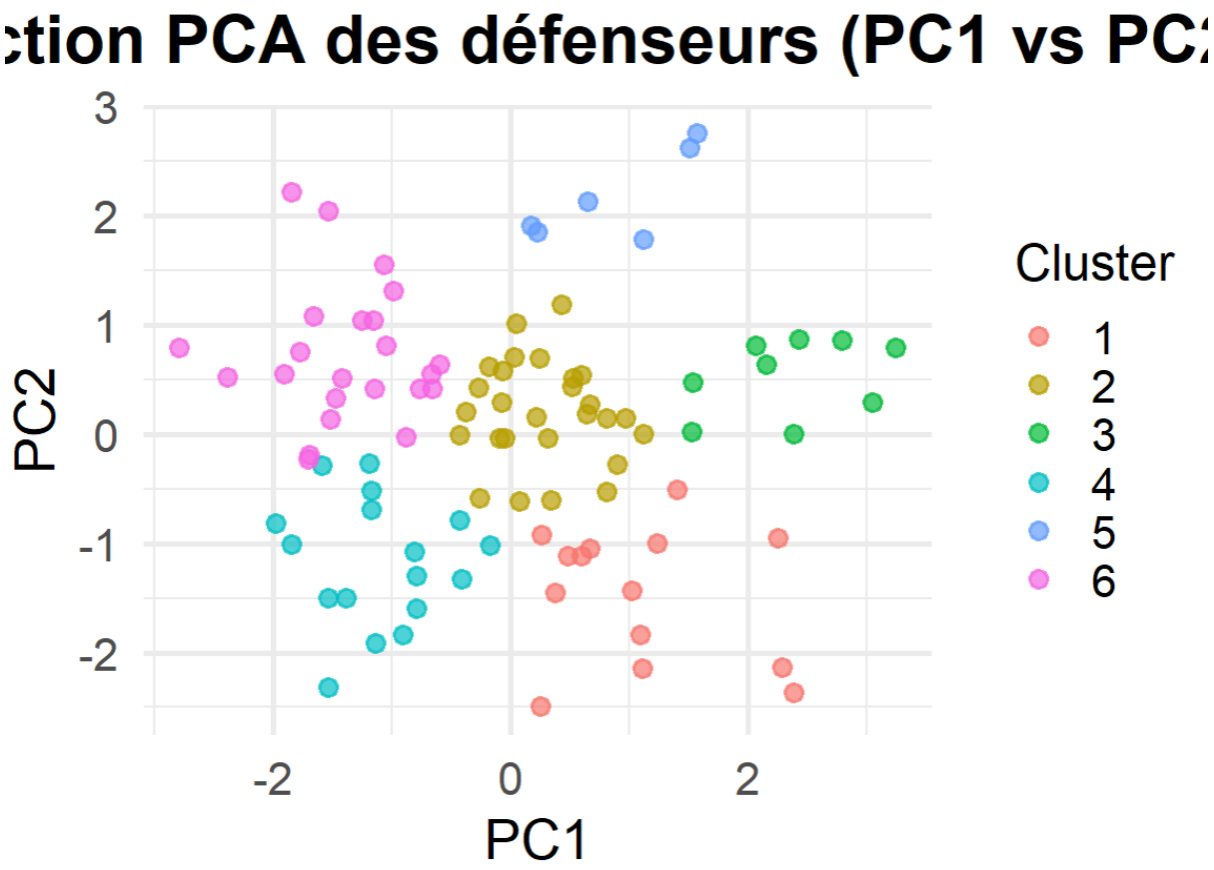
Players in this cluster show negative PC1 values but positive PC2 values. This suggests defenders who rely less on physical duels but contribute through positional awareness, interceptions, and controlled ball circulation. These players correspond to defenders who rely more on anticipation and reading of the game rather than direct confrontation.

Overall, the clustering analysis highlights the diversity of defensive roles within the dataset, ranging from physically dominant defenders to possession-oriented ball-playing defenders.

Methodological limitations:

It is important to note that clustering groups of players based on statistical similarity rather than tactical context, meaning that some players may appear in clusters that do not perfectly reflect their exact role within their team's defensive system.

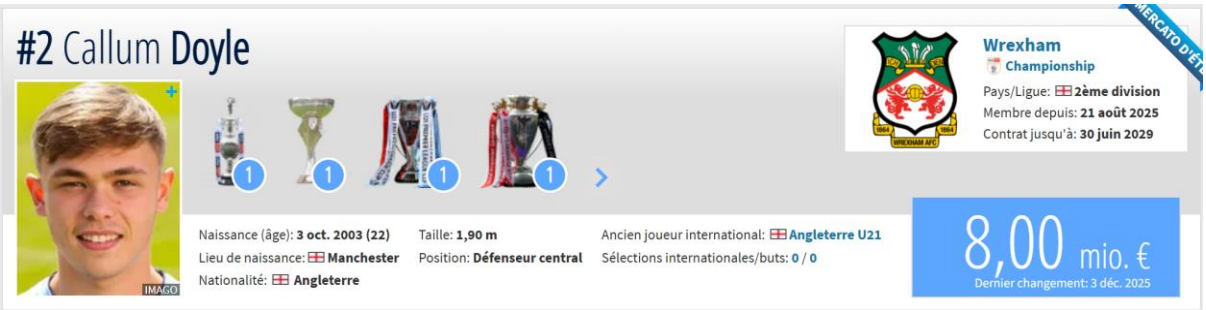
For example, the number of passes per90 is higher for center-backs because of their position (axial on the field so participate more to the buildup). So, when it comes to clustering some center-backs will be represented in cluster number five even if their offensive apport is low (key pass per90, assists per90...). The model captures more the ability of a player to circulate the ball than his ability to score/assist.



Example:

	player_name	source_league_name	age	score_defense	cluster
1	Jorge Salinas	Segunda División	18	0.85406680	1
2	C. Konaté	Ligue 2	21	0.29606148	1
3	Sergio Arribas	Segunda División	22	0.20458085	1
5	Z. Ariss	Ligue 2	21	1.37956976	2
6	Álex	Segunda División	22	1.04668854	2
7	A. Fontanarosa	Serie B	22	0.84899428	2
8	Naoufel El Hannach	Ligue 2	19	0.83956940	2
9	Amine Chabane	Ligue 2	19	1.20271570	3
10	G. Guarino	Serie B	21	0.16720723	3
13	C. Louis	Ligue 2	20	0.98184160	4
14	L. Villa	Serie B	22	0.28523614	4
15	A. Koné	Ligue 2	20	0.28187711	4
16	L. Noubi	Segunda División	20	0.21367599	4
17	B. Spencer	Championship	21	1.04157653	5
18	Unax Agote	Segunda División	22	0.87876060	5
21	C. Taylor	Championship	22	1.24410044	6
22	C. Doyle	Championship	22	0.82723018	6
23	Alessandro Dellavalle	Serie B	21	0.77090018	6
24	Tyler Bindon	Championship	20	0.66633116	6

Individual Player Analysis – Callum Doyle



Callum Doyle is a 22-year-old English central defender currently playing for Wrexham in the English Championship. During the 2025 season, he accumulated 1,885 minutes across 23 appearances, including 21 starts, indicating a stable and important role within the defensive line.

From a statistical perspective, Doyle stands out particularly for his strong involvement in ball circulation. He averages 56.6 passes per 90 minutes, significantly above the dataset average of approximately 46 passes per 90 minutes, reflected in a high standardized score ($z = 1.49$). This suggests a defender who actively contributes to build-up play and distribution from the back, a key characteristic of modern ball-playing central defenders.

Defensively, Doyle also shows consistent performance levels. He records 1.77 tackles per 90 minutes and 1.34 interceptions per 90 minutes, both close to or slightly above the dataset averages. In addition, he participates in 11.5 duels per 90 minutes and wins nearly 6, indicating regular involvement in defensive contests and aerial challenges.

Overall, Doyle combines defensive reliability with strong passing involvement, while showing moderate contribution in areas such as dribbling or attacking progression. His statistical profile therefore corresponds closely to that of a ball-

playing central defender, capable of contributing both to defensive stability and to possession phases through progressive distribution from the back.

Individual Player Analysis – Caleb Taylor

Caleb Taylor is a 22-year-old English defender currently playing for Millwall in the English Championship. During the 2025 season he accumulated 1,703 minutes across 22 appearances, including 18 starts, indicating a consistent presence in the defensive line.

From a defensive standpoint, Taylor shows strong involvement in anticipation and defensive recovery. He records 2.06 interceptions per 90 minutes, significantly above the dataset average of approximately 1.09, reflected in a very high standardized score (z = 2.97). In addition, he participates in 10.41 duels per 90 minutes and wins 7.24, highlighting strong effectiveness in defensive.

Taylor also demonstrates a notable level of offensive contribution for a central defender. He records 0.16 goals per 90 minutes and 0.11 assists per 90 minutes, both clearly above the dataset averages for defenders. His 1.11 shots per 90 minutes further suggest an active contribution during set pieces or attacking situations.

However, his involvement in ball circulation remains more moderate, with 37.15 passes per 90 minutes and 0.26 key passes per 90 minutes, indicating that his offensive impact is driven more by finishing actions and presence in attacking situations rather than by playmaking or build-up distribution.

Taylor's statistical profile therefore corresponds to that of a defensively proactive central defender with notable offensive efficiency, combining strong anticipation ability with an unexpected capacity to contribute to attacking phases.

Comparative Interpretation:

The comparison between Callum Doyle and Caleb Taylor illustrates two different defensive profiles identified within the dataset. Doyle appears as a distribution-oriented defender, strongly involved in the build-up of play and possession phases. In contrast, Taylor represents a more defensively proactive profile, characterized by strong interception ability and a higher contribution in attacking situations.

Clarification of the Clustering Structure:

Although six clusters were identified for defenders, these groups should not be interpreted as strict tactical roles, but rather statistical profiles based on similar defensive contributions.

Some clusters represent defenders strongly involved in physical defensive actions, particularly tackles and duels (high PC1 values), while others correspond to defenders contributing more through positional awareness and ball circulation, reflected by higher values in interceptions and passing activity (high PC2 values).

As with the other positions, the clustering therefore identifies patterns of statistical similarity rather than exact tactical functions, meaning that defenders with comparable roles in real matches may sometimes appear in different clusters depending on their statistical output.

Conclusion:

Despite certain limitations, the model provides a useful analytical framework for identifying defensive player profiles and comparing defenders across different leagues. By combining PCA and clustering techniques, the analysis highlights the diversity of defensive playing styles, ranging from physically dominant defenders to possession-oriented ball-playing profiles.

More broadly, this approach illustrates both the potential and the limits of data-driven player analysis, offering valuable insights into defensive performance while acknowledging that statistical models cannot fully capture the tactical context and responsibilities assigned to players within their teams.

Conclusion:

This study aimed to analyze and classify young professional football players based on their statistical performance using data-driven methods. By applying Principal Component Analysis and clustering techniques, the analysis identified several distinct performance profiles among attackers, midfielders, and defenders.

The results show that quantitative indicators can effectively capture important aspects of player behavior, such as finishing ability, creativity, defensive involvement, or ball circulation. Grouping players according to these statistical dimensions provides a useful framework for comparing players and identifying typical performance archetypes.

However, the analysis also highlights several limitations. Football performance is influenced by many contextual factors that are not fully captured by match statistics, such as tactical systems, positioning, off-ball movement, or team strategies. As a result, clustering methods identify statistical similarities rather than exact tactical roles.

Despite these limitations, the approach presented in this study offers a valuable tool for football analysis and scouting. By structuring complex performance data and identifying comparable player profiles across leagues, the model can support the evaluation of young talents and contribute to more informed recruitment decisions.

Future improvements could include the integration of additional variables, positional tracking data, or contextual information related to team tactics. Such developments would allow for a more comprehensive understanding of player performance and further enhance the potential of data-driven football analysis.